

Foundation (Jalapeno)

Q1. Factor the trinomial $x^2 + 5x + 6$.

- (A) $(x + 2)(x + 3)$
- (B) $(x + 1)(x + 4)$
- (C) $(x + 3)(x + 2)$
- (D) $(x + 4)(x + 1)$
- (E) $(x + 6)(x + 1)$

Q2. Factor the trinomial $x^2 - 7x + 12$.

- (A) $(x - 3)(x - 4)$
- (B) $(x - 1)(x - 6)$
- (C) $(x - 4)(x - 3)$
- (D) $(x - 2)(x - 5)$
- (E) $(x - 6)(x - 1)$

Q3. Factor the trinomial $x^2 + 2x - 15$.

- (A) $(x + 5)(x - 3)$
- (B) $(x + 1)(x - 15)$
- (C) $(x + 3)(x - 5)$
- (D) $(x - 1)(x + 15)$
- (E) $(x - 5)(x + 3)$

Q4. Factor the trinomial $x^2 - 9x + 20$.

- (A) $(x - 4)(x - 5)$
- (B) $(x - 1)(x - 8)$
- (C) $(x - 5)(x - 4)$
- (D) $(x - 2)(x - 7)$
- (E) $(x - 8)(x - 1)$

Q5. Factor the trinomial $x^2 + 10x + 24$.

- (A) $(x + 4)(x + 6)$
- (B) $(x + 1)(x + 9)$
- (C) $(x + 6)(x + 4)$
- (D) $(x + 2)(x + 8)$
- (E) $(x + 8)(x + 2)$

Q6. Factor the trinomial $x^2 - 2x - 15$.

- (A) $(x - 5)(x + 3)$
- (B) $(x - 1)(x + 15)$
- (C) $(x + 3)(x - 5)$
- (D) $(x + 1)(x - 15)$
- (E) $(x + 5)(x - 3)$

Q7. Factor the trinomial $x^2 + 4x - 21$.

- (A) $(x + 7)(x - 3)$
- (B) $(x + 1)(x - 21)$
- (C) $(x - 3)(x + 7)$
- (D) $(x - 1)(x + 21)$
- (E) $(x + 3)(x - 7)$

Q8. Factor the trinomial $x^2 - 3x - 10$.

- (A) $(x - 5)(x + 2)$
- (B) $(x - 1)(x + 10)$
- (C) $(x + 2)(x - 5)$
- (D) $(x + 1)(x - 10)$
- (E) $(x + 5)(x - 2)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $x^2 + 8x + 16$ and explain your steps.

Q10. Factor the trinomial $x^2 - 6x - 16$ and show your work.

Chef's Tips

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- Tip 1: Encourage students to use factoring by grouping or the quadratic formula if necessary.
- Tip 2: Remind students to check their factors by multiplying them to ensure the result equals the original trinomial.
- Tip 3: Suggest that students use the distributive property in reverse to factor trinomials.